

AWS Networking Cheat Sheet (Cloud Practitioner)

This cheat sheet summarizes the most important AWS networking concepts and exam tips for the AWS Cloud Practitioner exam.

Amazon VPC (Virtual Private Cloud)

- Logically isolated virtual network in AWS.
- You define IP ranges, subnets, route tables and gateways.
- Default VPC is automatically created in every AWS region.
- Key components: Subnets, Internet Gateway, NAT Gateway, Security Groups, NACL.

Subnets

- Divide a VPC into smaller networks.
- Public subnet has a route to Internet Gateway.
- Private subnet does not have direct internet access.
- Common design: Load balancer in public subnet, app servers in private subnet.

Internet Gateway (IGW)

- Enables communication between VPC and the internet.
- Must be attached to a VPC.
- Required for public subnet internet access.

NAT Gateway

- Allows private subnet instances to access the internet.
- Prevents inbound internet traffic.
- Highly available managed AWS service.

Security Groups vs Network ACL

- Security Groups operate at instance level.
- Network ACL operates at subnet level.
- Security Groups are stateful.
- Network ACLs are stateless.

Route Tables

- Control traffic routing inside VPC.
- Each subnet is associated with a route table.
- Routes specify destination and target (IGW, NAT, Peering).

VPC Peering

- Private connection between two VPCs.
- Traffic stays within AWS network.
- Works across accounts and regions.
- Connection is non-transitive.

AWS Transit Gateway

- Hub for connecting multiple VPCs.
- Supports hub-and-spoke architecture.
- Reduces complexity compared to many peerings.

VPN and Direct Connect

- Site-to-Site VPN: encrypted tunnel between on-premise and AWS.
- Client VPN: secure remote user access.
- Direct Connect: dedicated private fiber connection.

Route 53

- Highly available DNS service.
- Used for domain registration and traffic routing.
- Routing policies include Simple, Weighted, Latency, Failover.

Edge Networking Services

- CloudFront: CDN for low latency content delivery.
- Global Accelerator: improves TCP/UDP performance globally.
- S3 Transfer Acceleration: speeds up uploads to S3.

Important Exam Tips

- Security Groups = Stateful, NACL = Stateless.
- NAT Gateway = outbound internet for private subnet.
- Direct Connect = dedicated private connection.
- VPC Endpoint allows private access to AWS services.
- Transit Gateway connects multiple VPCs easily.